# Practice workbook · Explain how vaccination or infection gives active immunity

How vaccines or a first infection lead to antibodies and memory cells, why boosters are given, and herd protection.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VACCINATION-ACTIVE-IMMUNITY

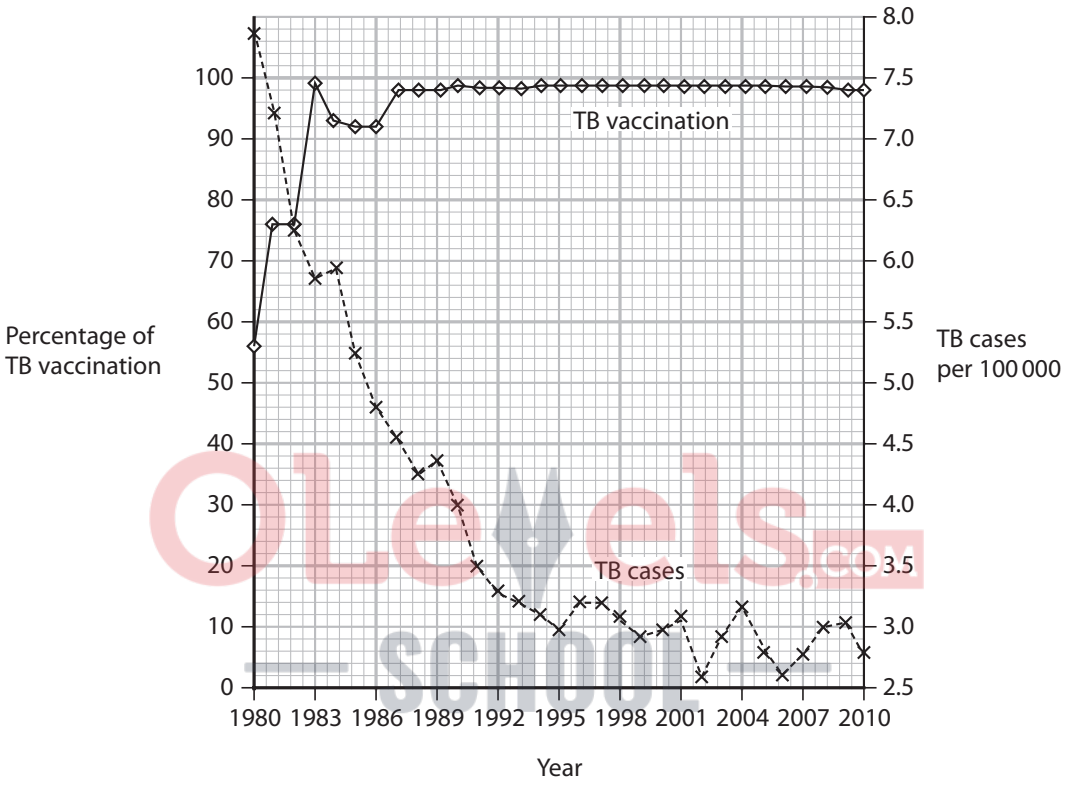
# 1. Question 4 · part c · 5 marks

June 2021 Paper 1

Worked-example retry · Attempt part c only.

**4** The graph shows the percentage of people vaccinated against tuberculosis (TB) in a country between 1980 and 2010.

It also shows the number of cases of tuberculosis in that country.




(a) (i) Tuberculosis is caused by the same type of organism as typhoid.

What is the type of organism?

**(1)**

- ☒ **A** bacterium
- ☒ **B** fungus
- ☒ **C** protozoan
- ☒ **D** virus

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# 1. Question 4 · part c · 5 marks

June 2021 Paper 1

Worked-example retry · Attempt part c only.

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(ii) Describe the relationship between the percentage vaccination and the number of cases of TB.

**(4)**

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(b) The population of this country in 2010 was 3.45 million.

Calculate the number of cases of tuberculosis in this country in 2010.

**(3)**

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number of cases = .....

# 1. Question 4 · part c · 5 marks

June 2021 Paper 1

Worked-example retry · Attempt part c only.

(c) Explain how vaccination can give a person immunity to a disease.

**(5)**

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**(Total for Question 4 = 13 marks)**

## 1. Question 4 · part c · 5 marks · Working and answer

Working and answer

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## 2. Question 8 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

**8** *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

**(3)**

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

| **Region of the world** | Q | R | S | T | U |
|---|---|---|---|---|---|
| **Total number of deaths from malaria** | 191 000 | 800 | 3800 | 14 400 | 1200 |
| **Number of deaths caused by *P. vivax*** | 1000 | 500 | 1400 | 4900 | 700 |

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

**(3)**

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## 2. Question 8 · part c · 3 marks

June 2019 Paper 1

Further practice · Attempt part c only.

(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

**(1)**

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

**(3)**

(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

**(3)**

**(Total for Question 8 = 13 marks)**

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## 2. Question 8 · part c · 3 marks · Working and answer

Working and answer

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## 3. Question 9 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

**9** (a) Hepatitis is a disease of the liver.

This disease can be caused by drinking too much alcohol.

Explain the damaging effects of alcohol on the liver.

**(3)**

(b) Hepatitis B is one type of virus that can cause hepatitis.

Describe the reproduction of a hepatitis B virus.

**(6)**

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## 3. Question 9 · part c · 3 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(c) Vaccinations can be used to prevent the spread of hepatitis.

Explain how vaccines can help people become immune to hepatitis.

**(3)**

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**(Total for Question 9 = 12 marks)**

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Working and answer

# 4. Question 3 · part b.iii · 2 marks

November 2023 Paper 2

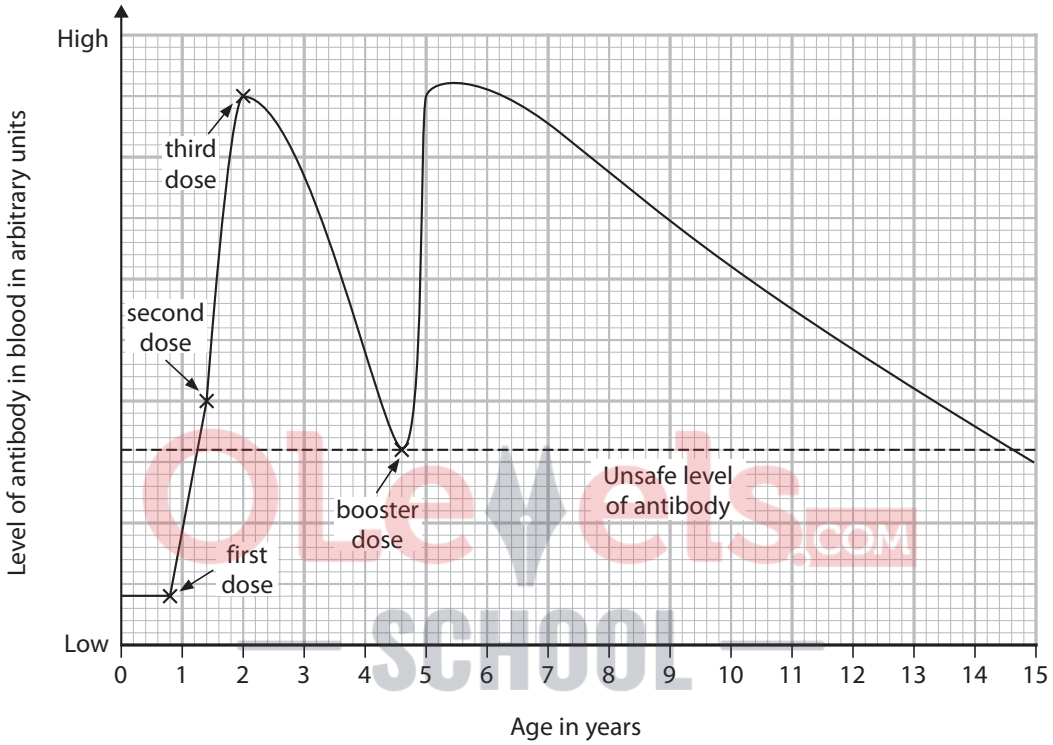
Further practice · Attempt part b.iii only.

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**3** A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.




(a) State the general name given to disease causing organisms.

**(1)**

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(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

**(1)**

time = ..... years

(ii) State the type of immunity that vaccination produces.

**(2)**

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# 4. Question 3 · part b.iii · 2 marks

November 2023 Paper 2

Further practice · Attempt part b.iii only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

**(2)**

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

**(2)**

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(d) Explain why a booster dose is necessary.

**(5)**

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## 4. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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# 5. Question 3 · part d · 5 marks

November 2023 Paper 2

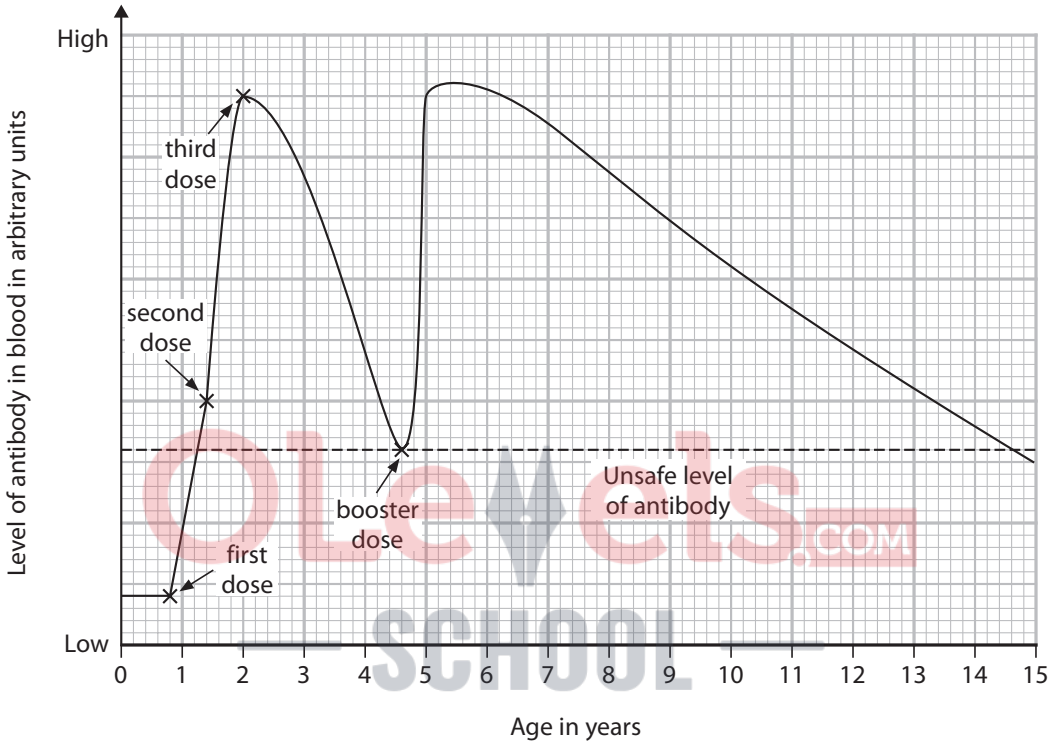
Further practice · Attempt part d only.

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**3** A vaccine is used to protect people against a disease.

The graph shows the level of antibodies in the blood of a child given the vaccine.

Three doses of the vaccine were given and then a booster dose was given.




(a) State the general name given to disease causing organisms.

**(1)**

.....

(b) (i) Determine the time, in years, between the first dose and the third dose of the vaccine.

**(1)**

time = ..... years

(ii) State the type of immunity that vaccination produces.

**(2)**

.....

.....

## 5. Question 3 · part d · 5 marks

November 2023 Paper 2

Further practice · Attempt part d only.

(iii) Name two components that could be in a vaccine that would help to trigger immunity.

**(2)**

1 .....

2 .....

(c) There are some antibodies in the blood of the child before the first vaccination is given.

Explain why there are antibodies in the blood before vaccination.

**(2)**

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(d) Explain why a booster dose is necessary.

**(5)**

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## 5. Question 3 · part d · 5 marks

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Further practice · Attempt part d only.

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(e) Give two differences between the effects of the first three doses and the effect of the booster dose.

**(2)**

1 .....

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2 .....

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**(Total for Question 3 = 15 marks)**

Working and answer

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# 6. Question 1 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

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**1** (a) The table shows some diseases.

Complete the table by writing in the type of pathogen that causes each disease.

**(3)**

| Disease | Type of pathogen |
|---|---|
| cholera | |
| AIDS | |
| athlete's foot | |

(b) The passage is about gonorrhoea.

Complete the passage using terms from the box.

**(4)**

| | | | |
|---|---|---|---|
| hormonal | antibiotics | contaminated water | virus |
| sexual intercourse | painkillers | bacterium | barrier |

The pathogen that causes gonorrhoea is a ..... .

The pathogen passes from one person to another by ..... .

Gonorrhoea can be prevented by using a ..... method of contraception.

A person with gonorrhoea can be prescribed ..... to kill the pathogen causing the disease.

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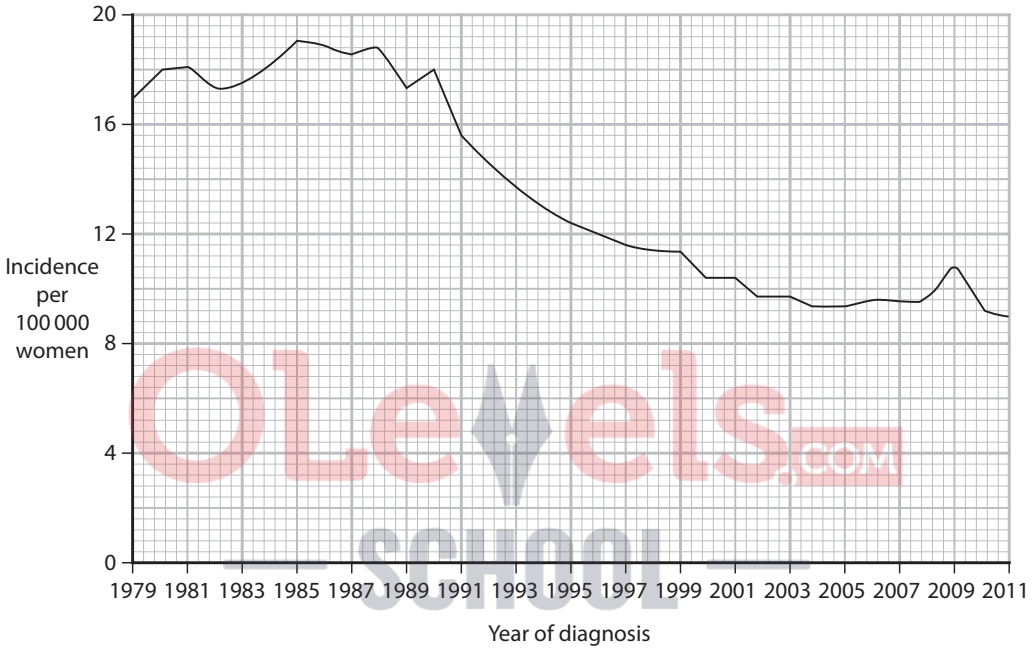
# 6. Question 1 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

(c) Cervical cancer is a disease that can be caused by a virus.

The graph shows the incidence of cervical cancer per 100 000 women in the United Kingdom between 1979 and 2011.




(Source: https://www.chattopadhyay.co.uk/cervical-cancer/)

(i) Describe the trends shown by the graph from 1979 to 2011.

**(2)**

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## 6. Question 1 · part d · 1 mark

June 2024 Paper 2

Further practice · Attempt part d only.

(ii) Calculate the difference in the incidence per 100 000 of women with cervical cancer between 1979 and 2011.

**(2)**

difference = ..... per 100 000

(d) Cervical cancer that is caused by a virus can now be prevented by vaccination.

The table gives information about vaccination.

Which row gives correct information about vaccination?

**(1)**

| | Contents of vaccination | Response by the body | How the pathogen is destroyed |
|---|---|---|---|
| ☒ **A** | antibodies | antigens produced | antigens destroy pathogen |
| ☒ **B** | antibodies | white blood cells produced | white blood cells destroy pathogen |
| ☒ **C** | antigens | antibodies produced | antibodies destroy pathogen |
| ☒ **D** | antigens | platelets produced | platelets destroy pathogen |

**(Total for Question 1 = 12 marks)**

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Working and answer

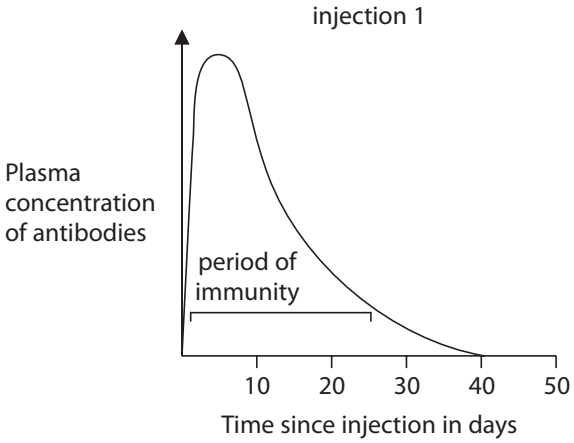
## 7. Question 4 · part b · 5 marks

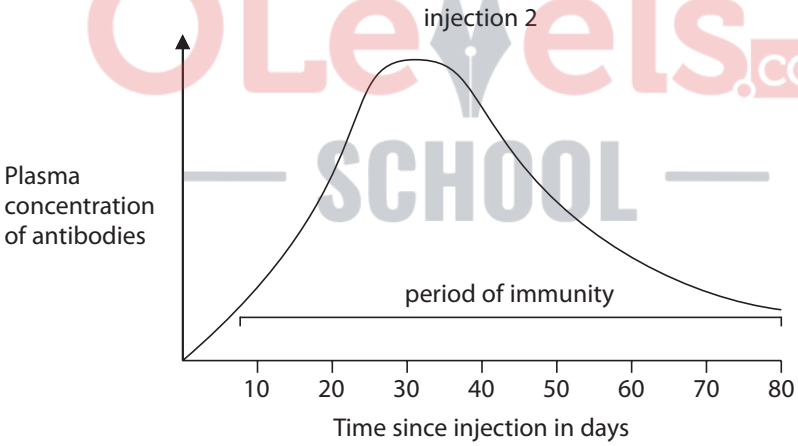
January 2022 Paper 2

Further practice · Attempt part b only.

**4** (a) Artificial immunity can be produced by different types of injection.

The graphs show the plasma concentration of antibodies produced as a result of two types of injection.




(i) Identify the different forms of artificial immunity produced by these two injections.

**(2)**

1 .....

2 .....

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# 7. Question 4 · part b · 5 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(ii) Explain which injection would be better for someone who already has the disease.

**(3)**

(b) Describe how natural immunity to a disease can happen.

**(5)**

**(Total for Question 4 = 10 marks)**

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## 7. Question 4 · part b · 5 marks · Working and answer

Working and answer

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